# Policy-Guided Schema-Aware Framework for Semi-Structured Document Information Extraction

Ayush Kumar
*School of Computer Science and Engineering (SCOPE)*
*Vellore Institute of Technology*
Vellore, India
ayush.kumar2022a@vitstudent.ac.in

Ramesh C
*School of Computer Science and Engineering (SCOPE)*
*Vellore Institute of Technology*
Vellore, India
ramesh.chandran@vit.ac.in

***Abstract*—Extracting typed fields from semi-structured documents requires values that are correct, conform to a schema and can be traced back to the page. We present a schema-aware extraction framework that runs an OCR-free Donut extractor and an OCR-grounded LLM extractor in parallel and reconciles them field by field with a deterministic, evidence-based fusion policy that records the source and the reason for every value. Donut answers are grounded in the OCR text before fusion, and field schemas are retrieved from an embedding-based registry. For layout analysis we train a lightweight LoRA adapter for LayoutLMv3 per document style; each adapter updates a small fraction of the parameters, occupies a few megabytes and clearly improves macro-F1 over a frozen backbone on both receipts and invoices and forms and multi-domain pages, with little variation across seeds. End to end, evaluated on SROIE, CORD and a synthetic multi-page contract set, the framework improves on its strongest single branch on CORD, exceeds a local vision-language model on the contract set and equals that model's exact-match rate on CORD, while the vision-language model remains strongest on SROIE. Grounding is the decisive fusion component and prevents the rise in hallucinated values that ungrounded OCR-free answers cause. Because every decision is logged, the remaining errors can be traced to individual policy rules.**

***Index Terms*—document information extraction, schema-aware extraction, LayoutLMv3, LoRA, semi-structured documents, policy-guided fusion**

## I. Introduction

Business processes produce large volumes of semi-structured documents such as receipts, invoices, forms and contracts. They combine natural-language text with layout structure, and the position of a field varies between versions, vendors and countries, so extraction rules that rely on fixed positions do not carry over.

Rule-based extraction was replaced by layout-aware models such as LayoutLM [1], LayoutLMv2 [2] and LayoutLMv3 [3], and later by OCR-free models such as Donut [4]. Large language models (LLMs) [5] and vision-language models (VLMs) [6] now extract fields from a schema given at run time, without task-specific training. Their outputs, however, are not anchored to the page [7]: a generator can return a well-formed value that the document does not contain, and a single model call gives no account of where a value came from.

We address this with a framework that treats extraction as a reconciliation problem. Two complementary extractors, an OCR-free Donut branch and an OCR-LLM branch that reads the OCR text and falls back to a VLM for missing fields, run in parallel. A deterministic policy then decides every field from evidence: whether a value is grounded in the OCR text, whether it parses as the field's type, whether it is reused across fields and whether the field is spatially anchored. Model confidence is never the deciding signal. The decision and its reason are logged per field, so every output value can be audited.

This paper makes four contributions. First, it introduces an evidence-based, field-level fusion policy that grounds OCR-free answers in the OCR text and stores an audit record for every field. Second, it trains style-specific LoRA adapters for LayoutLMv3 over a unified six-class layout label space, which improve test macro-F1 by up to 0.19 over a frozen backbone while updating 0.94% of the parameters. Third, it provides an embedding-based schema registry whose threshold rejects every unregistered document type in our query set, handing such documents to a schema agent instead of matching them to a wrong schema. Fourth, it evaluates the framework end to end on SROIE, CORD and a synthetic multi-page set against a local 7B VLM, with an ablation, a rule-level error attribution and a latency breakdown.

## II. Related Work

### A. Rule-Based and Template Extraction

Early document extraction systems relied on hand-written rules, regular expressions and templates keyed to fixed page positions. Declarative rule engines remain common in industry because their outputs are predictable and explainable [8], but every new vendor layout or document version requires new rules, and maintenance cost grows with the number of templates. Our fusion policy keeps the auditability of rules but applies them to the outputs of learned extractors rather than to raw page positions.

### B. Layout-Aware Pre-Trained Models

LayoutLM [1] adds two-dimensional position embeddings of OCR tokens to a BERT-style encoder, LayoutLMv2 [2] adds visual features and spatially aware self-attention, and LayoutLMv3 [3] unifies text and image masking with linear patch embeddings in place of a CNN backbone. BROS [9]

encodes the relative positions of text blocks, LiLT [10] decouples a language-independent layout stream so that it can be combined with different text encoders, and GeoLayoutLM [11] adds geometric pre-training aimed at relation extraction. FormNet [12] models label–value structure explicitly, using graph convolutions and rich attention to recover structure that is lost when a form is serialised.

These models are fine-tuned on benchmarks such as FUNSD [13] for forms and CORD [14] and SROIE [15] for receipts, and region-level corpora such as DocLayNet [16] supply labels for layout analysis. All of them depend on an OCR engine such as Tesseract [17] and inherit its errors, and each new schema needs a task-specific head and labelled data.

### C. OCR-Free and Generative Document Models

OCR-free models read the page image directly and generate their output as text. Donut [4] pairs a Swin Transformer encoder with a BART decoder and is pre-trained to read text before being fine-tuned for parsing or question answering, and Pix2Struct [18] is pre-trained to parse masked web-page screenshots into simplified HTML. Because their answers are generated rather than located, these models avoid OCR error propagation but can return text that does not appear on the page, and the token probabilities of modern neural networks are generally not calibrated to correctness [19]. Posing one question per field to a model fine-tuned on DocVQA [20] makes the output schema-driven without retraining; we use Donut in this way and check each answer against the OCR text before it can be used.

### D. LLM and VLM Extraction

Large language models (LLMs) extract entities and fields from a prompt with little or no annotation; GPT-NER [5], for instance, recasts sequence labelling as text generation. For documents, the OCR text is serialised into the prompt, which discards most of the layout and limits the length that can be processed, so long multi-page documents remain difficult; CUAD [21] provides clause-level annotations for contracts and Kleister [22] key-information labels for long documents with complex layouts. Generative extractors can also produce fluent, well-formed values that the source does not support [7]. Retrieval-augmented generation [23] reduces this by conditioning the generator on retrieved evidence, but it does not check after generation that the answer is actually supported.

Agentic methods such as ReAct [24] let a model interleave reasoning with actions and revise intermediate results. They can correct errors, but the result depends on a sequence of sampled model calls and the reason for a final value is hard to reconstruct. Vision-language models (VLMs) such as Qwen2.5-VL [6] read the page image together with a requested output schema and are strong single-call extractors on receipts. A single call, however, returns one value per field without a record of where it came from or whether another reading of the page disagrees. We use a local VLM both as a baseline and as the vision fallback inside the OCR-LLM branch.

### E. Combining Multiple Extractors

Combining the outputs of several systems is well established. ROVER [25], for example, aligns the outputs of several speech recognisers and votes word by word, optionally weighted by confidence. Such schemes assume members of comparable quality or calibrated scores. Neither holds for an OCR-free generator and an OCR-text extractor, whose errors differ in kind and whose confidences are not comparable [19]. Research on data provenance distinguishes why a value appears in an output from where it was copied from [26]; this distinction motivates our per-field audit record, which stores for every fused value the branch it came from and the rule that selected it.

### F. Schema Matching and Retrieval

Matching an incoming set of fields to a known schema is a classical database problem. Rahm and Bernstein [27] classify matchers by whether they use element names, instances or structure, and by how several matchers are combined. Sentence embeddings such as Sentence-BERT [28] allow the similarity of field names and descriptions to be computed as a cosine in a shared space, which suits schemas whose field names vary between sources. Document extraction systems, however, usually assume one fixed schema per model, and schema matching is rarely evaluated for its ability to reject document types that are not registered.

### G. Parameter-Efficient Adaptation

Adapters [29] insert small trainable bottleneck layers into a frozen network, and LoRA [30] learns low-rank updates to the attention weights that can be merged into them after training. Both have been studied mainly for language models. Their use for layout-aware document encoders across document styles with heterogeneous label sets, where one frozen base serves several style-specific adapters, has received little study.

## III. Identified Gaps and Problem Statement

### A. Identified Gaps

Layout-aware, OCR-free and generative extractors have been studied largely in isolation, and where outputs are combined, voting or confidence weighting presupposes calibrated scores that OCR-free generators do not provide. Verifying OCR-free answers against the OCR text after generation, and recording for each field which component produced its value and why, have not been reported. Extraction pipelines also rarely resolve schemas at run time while rejecting unregistered document types, or compare themselves with a single VLM call on the page.

### B. Problem Statement

Given a PDF $D$, turned by ingestion into OCR or native-text blocks $B$, and a schema $S = \{f_1, \ldots, f_n\}$ of typed fields with constraints, supplied or resolved from a registry $\mathcal{R}$, produce for every field a value $v_i$ (or null), a confidence $c_i \in [0, 1]$ and an audit record $a_i$ naming the source branch and selecting rule. Each $v_i$ must match the document, satisfy

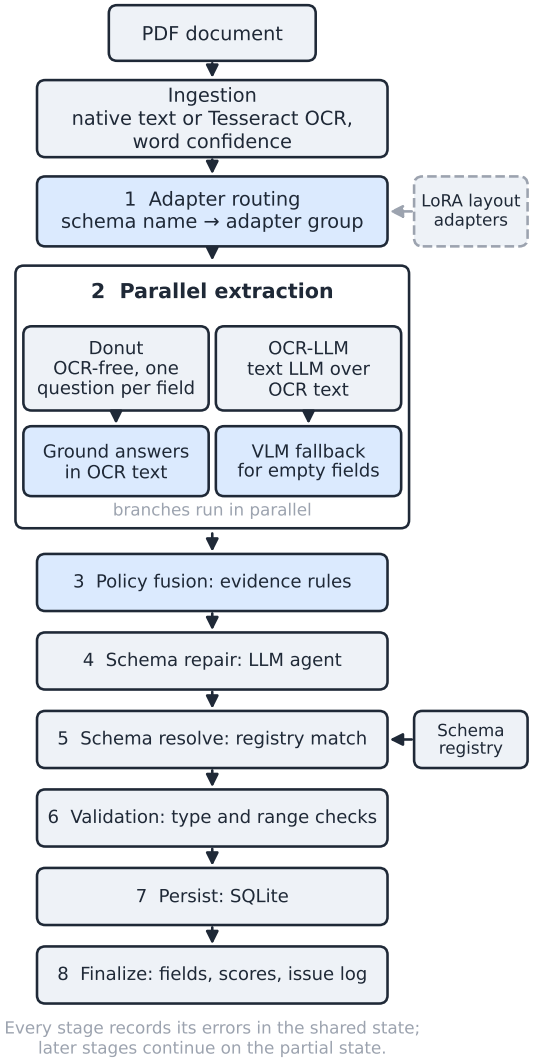



Fig. 1. Pipeline architecture: ingestion, the eight orchestrator stages and the two parallel extraction branches.

its constraints or be flagged, be supported by $B$ or an explicit vision fallback, and be a deterministic function of the branch outputs; an unregistered document type must not be matched to a wrong schema.

## IV. System Architecture and Methodology

**Orchestration.** Fig. 1 shows the eight stages. Ingestion feeds stage 1, which selects a LoRA layout adapter (dashed); the two extraction branches of stage 2 run in parallel and are reconciled in stage 3. The stages share a state holding the document blocks, the rendered page images, the selected adapter, the extraction bundle of both branches, the fused fields and their audit records, the resolved schema, the validation outcomes, the storage key and any accumulated errors. Each stage handles its own exceptions, so a failure in one stage leaves a partial state that later stages can still use. The adapter group is selected from the identified or retrieved schema name; the adapters label layout regions, and the field extractors take spatial information directly from the OCR bounding boxes.

**Algorithm 1:** Deterministic field-level policy fusion

**Input:** OCR-LLM value $v_O$, confidence $c_O$; grounded Donut value $v_D$, confidence $c_D$; floor $\delta$; field type $\tau$
**if** $v_O$ and $v_D$ are both null **then**
  **return** null
**else if** $v_D$ is null **then**
  **return** $v_O$ if $c_O \geq \delta$, else null
**else if** $v_O$ is null **then**
  **return** $v_D$ if $c_D \geq \delta$, $v_D$ is usable and $\tau$ is not numeric; else null
**else if** canonical($v_O$) = canonical($v_D$) **then**
  **return** $v_O$ {agreement}
**else if** field is spatially anchored **then**
  **return** $v_O$
**else if** $v_O$ is unusable **and** $v_D$ is usable **then**
  **return** $v_D$ {evidence, not confidence}
**else**
  **return** $v_O$ {default}
**end if**

### A. Document Ingestion with OCR Fallback

Ingestion turns a PDF into blocks, each with page number, text, bounding box $(x_0, y_0, x_1, y_1)$, confidence and a source tag (native or OCR). Native text is read with PyMuPDF; a page with fewer than 20 characters or five words is rendered at $2\times$ scale (halved above 5000 pixels) and passed to Tesseract, whose word-level confidences are kept for the fusion layer.

### B. Parallel Extraction

The Donut and OCR-LLM extractors run on two worker threads and return, per field, a value, a confidence and spatial metadata.

**Donut branch.** Each field $f_i$ is posed as a question $q_i$ derived from its description (e.g., "What is the invoice date?"), giving answer$_i$ = Donut(image, $q_i$) with confidence

$$c_i^{\text{donut}} = \frac{1}{T}\sum_{t=1}^{T} \max_v \text{softmax}(s_t)_v, \tag{1}$$

where $s_t$ is the logit vector at generation step $t$ of $T$. Weights are loaded lazily on first use.

**OCR-LLM branch.** A text LLM receives the first 4000 characters of the document and all field descriptions and returns a JSON extraction. Each field left empty is then passed with the page image to a VLM, which recovers values that depend on the image, such as stamps, tables and signatures. Extracted values are matched to OCR blocks to obtain their bounding boxes.

### C. Evidence-Based Policy Fusion

Donut's token probabilities are poorly calibrated: we observed 0.9–1.0 for both correct and incorrect answers. The fusion layer therefore decides from evidence and uses confidence only as a floor $\delta$ below which a lone value is discarded.

**Grounding.** Each Donut answer is searched in the OCR blocks of its page. An answer found in no block is dropped; a supported answer takes the confidence of the block that contains it.

**Usability.** A value is unusable if it does not parse as its field type (a date without a four-digit year, a non-numeric amount), if the same value was assigned to several fields, or if it is a clause of more than ten words rather than a value.

**Decision.** The OCR-LLM value is the default, and Donut takes over only when the OCR-LLM value is unusable and its own value is usable. Spatially anchored fields (dates, amounts, signatures) keep the OCR-LLM value on conflict. A Donut value may fill a field the OCR-LLM branch left empty only for non-numeric fields, because Donut-only numeric fills were the main source of wrong values in development. Algorithm 1 gives the per-field procedure.

Fusion also normalizes values (whitespace, punctuation and boilerplate such as "as defined herein") and runs cross-field checks such as effective_date $\leq$ expiry_date and disclosing_party $\neq$ receiving_party. Failed checks are flagged in the record metadata, not silently corrected. Each fused field is stored with its source branch and the rule that selected it.

### D. LoRA Adapter-Routed Layout Analysis

**Dataset grouping.** Four datasets with token- or region-level layout annotations form two groups by document style: Group 1 (receipts and invoices: CORD and SROIE, 1,363 training pages) and Group 2 (forms and multi-domain pages: FUNSD and DocLayNet, 4,954 training pages with DocLayNet capped at 5,000). Their layout statistics differ, so each group gets its own adapter on a shared frozen LayoutLMv3-base. A router keeps the base model and a few recently used adapters in memory and switches between them with the PEFT calls `load_adapter` and `set_adapter`. The adapters predict layout classes, not field values, so their training data does not leak extraction answers into the end-to-end evaluation.

**Unified label space.** The datasets differ in granularity: word-level entities in FUNSD, region categories in DocLayNet and receipt fields in CORD and SROIE. All are mapped to six labels: paragraph, heading, list item, table, caption and other. Each OCR token, with its box normalized to $[0, 1000]$, receives the label of the first annotated segment that contains its centre point, or the background label otherwise. This turns word-, region- and document-level supervision into one token-level task and makes the two adapters directly comparable: because they share the base model and the label set, either adapter can label any document without retraining, which the cross-group evaluation uses.

**Adapter configuration.** LoRA is applied to the query, key, value and output projections with rank $r = 16$, $\alpha = 32$ and dropout 0.05. Training uses AdamW (learning rate $5 \times 10^{-5}$, weight decay 0.01), a cosine schedule with 6% warmup, mini-batch 2 with 8 accumulation steps (effective batch 16), 512-token inputs, bf16 precision, at most 10 epochs and early stopping on validation macro-F1 with patience 3. Every run writes its full configuration, per-epoch metrics and test metrics to JSON files, from which all layout-adapter tables and figures are generated.

### E. Schema Registry, Validation and Storage

Each schema is stored as the averaged all-MiniLM-L6-v2 embedding of its field descriptions and retrieved by cosine similarity with threshold $\sigma$. Below the threshold, a schema agent (GPT-OSS-120B/20B) matches fields, normalizes names or synthesizes a new schema; the same agent repairs fields that fail validation. Validation checks type and value constraints; an invalid field is retried or set to its schema default. Valid records are stored in SQLite.

## V. Experimental Results and Evaluation

### A. Experimental Setup

All experiments ran on a Windows 11 laptop with Python 3.12, PyTorch and one NVIDIA RTX 5060 Laptop GPU (8 GB). Orchestration uses LangGraph and LangChain, fine-tuning uses PEFT, and storage uses SQLite.

**Layout adapters.** Each configuration was trained with seeds 13, 42 and 1234 on a fixed train/validation split: the datasets' own validation splits where they exist (CORD, DocLayNet) and a seeded 10% hold-out otherwise (SROIE, FUNSD). Official test splits were used only for final evaluation: CORD (100) and SROIE (347) pages for Group 1, and FUNSD (50) and the first 834 DocLayNet test pages for Group 2. The baseline freezes the whole backbone and trains only the token-classification head, with the same data, seeds, schedule and early stopping.

**End-to-end extraction.** We evaluate on a seeded random subset of 100 SROIE test receipts (company, date, address, total), a seeded subset of 50 CORD test receipts (six amount fields) and a 15-document synthetic multi-page set (invoices, NDAs, employment contracts, land deeds and tender instruments; 30 fields). Subsets are drawn once from the official test splits with subset seed 0, and no setting was tuned on them. All model stages run locally at temperature 0: Donut-base fine-tuned on DocVQA, Llama 3.2 3B for the text pass, and Qwen2.5-VL-7B served by Ollama (8192-token context) for the vision fallback and as the VLM-only baseline. F1 is field-level token-overlap F1 after normalization, and EM is the share of gold values matched exactly. Code, result files and figure scripts are available at github.com/AyushK0808/multiagent-form-schema-etl.

### B. Layout Adapters

Table I shows that LoRA improves test macro-F1 over the frozen-backbone head by 0.068 for Group 1 and 0.189 for Group 2, while updating 0.94% of the parameters, and the improvement is far larger than the seed variation (standard

TABLE I
LAYOUT LABELLING MACRO-F1 (MEAN ± STD, THREE SEEDS).

| Method | Grp. | Train. | Val | Test | Cross |
|---|---|---|---|---|---|
| Frozen | 1 | 0.0037% | 0.715 ± .007 | 0.706 ± .004 | – |
| LoRA | 1 | 0.94% | **0.807** ± .008 | **0.774** ± .008 | 0.195 ± .037 |
| Frozen‡ | 2 | 0.0037% | 0.542 | 0.550 | – |
| LoRA | 2 | 0.94% | **0.755** ± .006 | **0.739** ± .005 | 0.128 ± .005 |

Frozen: frozen backbone, trained head. Train.: share of the 126.5M parameters updated. Cross: macro-F1 on the other group's test set. ‡One seed.

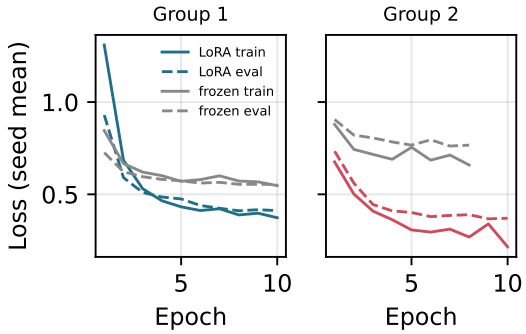



Fig. 2. Training (solid) and validation (dashed) loss per epoch (seed means).

deviation at most 0.008). Test accuracy is 0.851 and 0.898. Fig. 3 shows where the improvement comes from (seed means; Group 1 has no table or caption class): it is largest on the classes the frozen head handles worst, Group 2 list items (0.081 to 0.574) and captions (0.294 to 0.527), and Group 1 headings (0.70 to 0.87). Paragraphs (0.90–0.94) and tables (0.93) are recognised reliably. Group 2 is dominated by DocLayNet; on its 50 FUNSD forms the adapter reaches 0.22 macro-F1, because the form labels map unevenly onto the shared label space. Applied to the other group's test set, macro-F1 falls to 0.195 and 0.128. The adapters are therefore specialised, which supports routing each document to the adapter for its style.

Each adapter has 1.18M trainable parameters and occupies 4.5 MB, against about 500 MB for a full LayoutLMv3 checkpoint, so many document styles can share one base model. Training took about 260 s per epoch for Group 1 and 1,000 s for Group 2 with a peak GPU memory of 2.9 GB, and validation macro-F1 came within about 0.05 of its final value after four to five epochs. Fig. 2 shows that the adapters reach a clearly lower validation loss than the frozen head, which plateaus early. The gain therefore comes from added capacity in the encoder, not from longer training of the head.

### C. End-to-End Extraction

Table II gives the main results. On the synthetic multi-page set, the closest to the contract use case that motivated the design, the full pipeline reaches 0.980 F1 and 0.973 EM, above the VLM (0.973 and 0.960). On CORD the fusion adds 0.018 F1 and 0.029 EM to its OCR-LLM branch, reaching the VLM's exact-match rate and recall (both 0.900) and coming

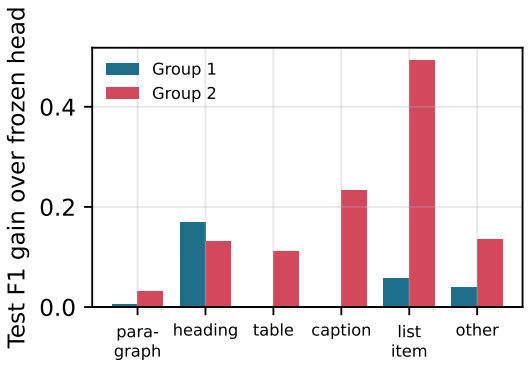



Fig. 3. Per-class test F1 gain of LoRA over the frozen head.

TABLE II
END-TO-END TOKEN-OVERLAP F1 AND EXACT MATCH (EM).

| System | Synthetic (15) F1 | Synthetic (15) EM | SROIE (100) F1 | SROIE (100) EM | CORD (50) F1 | CORD (50) EM |
|---|---|---|---|---|---|---|
| VLM only | 0.973 | 0.960 | **0.992** | **0.860** | **0.843** | **0.900** |
| OCR-LLM branch | **0.980** | **0.973** | 0.816 | 0.557 | 0.818 | 0.871 |
| Donut only† | 0.257 | 0.120 | 0.208 | 0.107 | 0.000 | 0.000 |
| Text pass only† | 0.673 | 0.507 | 0.666 | 0.367 | 0.023 | 0.012 |
| Full pipeline | **0.980** | **0.973** | 0.801 | 0.547 | 0.836 | **0.900** |

VLM only: Qwen2.5-VL-7B on the page image and schema. Text pass: text-only first pass of the OCR-LLM branch. †Limited by sparse OCR on CORD.

within 0.007 F1 of it. On SROIE the VLM is the strongest system (0.992), and the pipeline stays within 0.015 F1 of its OCR-LLM branch. Neither specialist branch is competitive alone: Donut reaches at most 0.257 F1 and the text pass at most 0.673, whereas the OCR-LLM branch with its vision fallback and the fused pipeline reach 0.80–0.98.

The VLM is also part of the framework, as the vision fallback of the OCR-LLM branch. Its strength on SROIE, whose four fields are printed prominently on a single page, shows how much the vision path can contribute, and the evidence rules in Algorithm 1 extend directly to the VLM answer as a third source.

### D. Ablation

On the synthetic set every variant equals the full pipeline (0.980), so Table III reports SROIE and CORD. It identifies grounding as the decisive component: removing it costs 0.031 F1 on SROIE and 0.011 on CORD, and on SROIE it lowers the median per-document F1 from 0.850 to 0.788 and nearly doubles the share of receipts scoring 0.5 or less (8% to 14%). Restricting Donut fills to non-numeric fields matters as well: letting Donut fill every empty field lowers CORD F1 to 0.810 and raises the hallucination rate from 0.148 to 0.192. This default was fixed before the test runs. Schema-agent repair, normalization and the spatial rule each change F1 by at most 0.002 on these short receipts. They are safeguards for long documents, where normalization and cross-field checks have more to act on.

TABLE III
ABLATION (TOKEN-OVERLAP F1; CORD HALLUCINATION RATE IN PARENTHESES).

| Variant | SROIE | CORD |
|---|---|---|
| Full pipeline | **0.801** | **0.836** (0.148) |
| without grounding of Donut answers | 0.770 | 0.825 (0.148) |
| without schema-agent repair | 0.799 | 0.836 (0.148) |
| without value normalization | 0.799 | 0.836 (0.148) |
| without spatial-field rule | 0.799 | 0.836 (0.148) |
| Donut fills all empty fields | 0.801 | 0.810 (0.192) |
| Donut fills no empty fields | 0.801 | 0.836 (0.148) |

TABLE IV
PER-FIELD TOKEN-OVERLAP F1.

| | SROIE (100) | | | | CORD (50) | | |
|---|---|---|---|---|---|---|---|
| **Field** | VLM | OCR-LLM | Full | **Field** | VLM | OCR-LLM | Full |
| company | 0.99 | 0.76 | 0.77 | subtotal | 0.78 | 0.75 | 0.78 |
| date | 0.99 | 0.92 | 0.92 | tax | 0.74 | 0.71 | 0.74 |
| address | 0.99 | 0.84 | 0.76 | total | 0.86 | 0.84 | 0.86 |
| total | 1.00 | 0.75 | 0.76 | cash | 0.87 | 0.84 | 0.87 |
| | | | | change | 0.91 | 0.90 | 0.90 |
| | | | | credit c. | 1.00 | 0.89 | 0.89 |

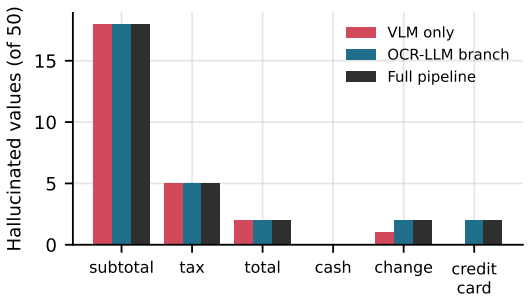



Fig. 4. CORD: predicted values where the gold value is empty.

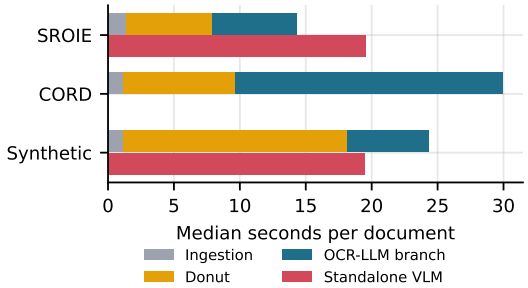



Fig. 5. Median seconds per document by stage, against a standalone VLM call.

### E. Audit Trail and Error Attribution

Because every fused value records the rule that selected it, the differences between the pipeline and its OCR-LLM branch can be attributed exactly. Table IV gives per-field F1, pooled over documents as in Table II. On CORD the fused pipeline is at or above its branch on every field and equals the VLM on subtotal, tax, total and cash. There the policy changes six values: five become correct (one corrected cash amount and four amounts filled in) and none becomes wrong. On SROIE it changes 21 of 400 values. Six become better (three company names, two totals, one address) and one is unchanged in score. The other 14 are all addresses, and every one of the 15 address overrides is triggered by the same rule: Malaysian addresses often exceed ten words, so the clause test marks the correct OCR-LLM value as unusable and a shorter Donut fragment replaces it. The SROIE gap between the pipeline and its branch is thus caused by a single rule, which the log identifies directly; restricting the clause test to clause-type fields is a one-line policy change. We report the numbers obtained with the policy as fixed before testing.

Fig. 4 breaks down the CORD hallucination rate, which is similar for all systems (0.148 for the pipeline, 0.151 for the OCR-LLM branch and 0.135 for the VLM). It is an annotation effect. Subtotal accounts for 18 values in every system, tax for five and total for two, and change and credit-card amounts add one value for the VLM and four for the other two systems. For the VLM, 16 of its 18 subtotals copy another gold field of the same receipt, usually the total, on receipts that print no subtotal line. Excluding subtotals, the VLM's rate falls to 5.6%. CORD also shows the dependence of grounding on OCR coverage: Tesseract returns about 11 text blocks per CORD receipt, against 29.5 on SROIE, so grounding removes nearly all of Donut's answers (about six per receipt before grounding), and the Donut-only and text-pass rows of Table II measure OCR coverage rather than those models. A stronger OCR engine directly increases what grounding can retain.

### F. Latency

We timed 15 documents (five per dataset, after one discarded warm-up) with response caching disabled, running the stages one after the other; Fig. 5 reports medians. CORD VLM timings are omitted because Ollama answered them from its prompt cache. Fusion takes under 3 ms and OCR about 1.2 s. On SROIE, where the text pass fills most fields, the whole pipeline needs 14.3 s per receipt against 19.5 s for a standalone VLM call. On CORD, where sparse OCR forces the vision fallback, the pipeline median is 29.7 s, and on the multi-page synthetic documents 24.4 s, of which Donut accounts for 17 s. Running the two branches concurrently, as in deployment, bounds the extraction time by the slower branch instead of their sum.

### G. Schema Registry Retrieval

The registry holds eight schemas: the four synthetic document types, SROIE and CORD receipts, purchase orders and forms. The query set has 50 labelled field-name sets. Forty belong to a registered schema (exact names, synonym renames, subsets, and subsets with unrelated extra fields; five per schema), and ten describe unregistered document types and must be rejected. At the operating threshold $\sigma = 0.70$ (Fig. 6), precision is 1.00 and no unregistered query is accepted, with recall 0.575; queries below the threshold go to the schema

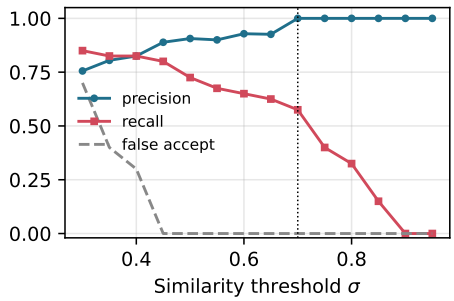



Fig. 6. Schema retrieval precision and recall against threshold $\sigma$.

agent instead of being matched to the wrong schema. Precision stays high over a wide range: at $\sigma = 0.45$ it is 0.889 at recall 0.800, still with no false acceptance, and false acceptances begin only at $\sigma = 0.40$. A deployment that prefers fewer agent calls can therefore lower $\sigma$ substantially without accepting the wrong schema.

## VI. Conclusion

We presented a schema-aware framework that reconciles an OCR-free and an OCR-grounded extractor field by field with a deterministic, evidence-based policy, together with an embedding-based schema registry and style-specific LoRA layout adapters. The adapters improve test macro-F1 over a frozen backbone to 0.774 and 0.739 at 0.94% trainable parameters and 4.5 MB each, and they are specialised enough to justify routing documents by style. End to end, the pipeline reaches 0.980 F1 on a multi-page synthetic set, above a 7B VLM; it improves on its strongest branch on CORD and matches the VLM's exact-match rate there; and grounding OCR-free answers in the page text is its most effective component and keeps hallucinated values in check. At its operating threshold the schema registry accepts no unregistered document type. Finally, the per-field audit record turns error analysis into rule analysis: the gap between the pipeline and its branch on SROIE traces to a single clause-length rule, which the log identifies without further experiments.

## VII. Limitations

The public benchmarks are receipt subsets of 100 and 50 documents evaluated in one run per configuration, so differences under 0.02 F1 are not resolved, and the multi-page set is small, synthetic and written by the authors. Grounding can keep only what the OCR engine reads, which limits it on CORD, where Tesseract output is sparse. The policy rules and thresholds were set by hand on development documents and may need revisiting for a new domain, as the clause test on long addresses shows. Finally, the layout adapters are evaluated as a standalone component and are not yet used by the field extractors.

## VIII. Future Work

The VLM, now a fallback inside the OCR-LLM branch, can enter the policy as a third evidence source, subject to the same grounding and usability checks. Restricting the clause test to clause-type fields would remove the SROIE regression that the audit log attributes to it. The region labels of the layout adapters can be passed to the field extractors, for example to restrict grounding to the region a field belongs to, and a stronger OCR engine would let grounding retain more of Donut's output. Finally, we plan to evaluate on long multi-page documents with public annotations, such as Kleister [22] and CUAD [21].

## References

[1] Y. Xu, M. Li, L. Cui, S. Huang, F. Wei, and M. Zhou, "LayoutLM: Pre-training of text and layout for document image understanding," in *Proc. 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2020, pp. 1192–1200.

[2] Y. Xu, Y. Xu, T. Lv, L. Cui, F. Wei, G. Wang, Y. Lu, D. Florencio, C. Zhang, W. Che, M. Zhang, and L. Zhou, "LayoutLMv2: Multi-modal pre-training for visually-rich document understanding," in *Proc. 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP)*, 2021, pp. 2579–2591.

[3] Y. Huang, T. Lv, L. Cui, Y. Lu, and F. Wei, "LayoutLMv3: Pre-training for document AI with unified text and image masking," in *Proc. 30th ACM International Conference on Multimedia (ACM MM)*, 2022, pp. 4083–4091.

[4] G. Kim, T. Hong, M. Yim, J. Nam, J. Park, J. Yim, W. Hwang, S. Yun, D. Han, and S. Park, "OCR-free document understanding transformer," in *Proc. European Conference on Computer Vision (ECCV)*, 2022, pp. 498–517.

[5] S. Wang, X. Sun, X. Li, R. Ouyang, F. Wu, T. Zhang, J. Li, G. Wang, and C. Guo, "GPT-NER: Named entity recognition via large language models," in *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025, pp. 4257–4275.

[6] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang, H. Zhong, Y. Zhu, M. Yang, Z. Li, J. Wan, P. Wang, W. Ding, Z. Fu, Y. Xu, J. Ye, X. Zhang, T. Xie, Z. Cheng, H. Zhang, Z. Yang, H. Xu, and J. Lin, "Qwen2.5-VL technical report," *arXiv preprint arXiv:2502.13923*, 2025.

[7] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, Art. 248, pp. 1–38, 2023.

[8] L. Chiticariu, Y. Li, and F. R. Reiss, "Rule-based information extraction is dead! Long live rule-based information extraction systems!" in *Proc. 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2013, pp. 827–832.

[9] T. Hong, D. Kim, M. Ji, W. Hwang, D. Nam, and S. Park, "BROS: A pre-trained language model focusing on text and layout for better key information extraction from documents," in *Proc. AAAI Conference on Artificial Intelligence*, vol. 36, no. 10, 2022, pp. 10767–10775.

[10] J. Wang, L. Jin, and K. Ding, "LiLT: A simple yet effective language-independent layout transformer for structured document understanding," in *Proc. 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, pp. 7747–7757.

[11] C. Luo, C. Cheng, Q. Zheng, and C. Yao, "GeoLayoutLM: Geometric pre-training for visual information extraction," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 7092–7101.

[12] C.-Y. Lee, C.-L. Li, T. Dozat, V. Perot, G. Su, N. Hua, J. Ainslie, R. Wang, Y. Fujii, and T. Pfister, "FormNet: Structural encoding beyond sequential modeling in form document information extraction," in *Proc. 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, pp. 3735–3754.

[13] G. Jaume, H. K. Ekenel, and J.-P. Thiran, "FUNSD: A dataset for form understanding in noisy scanned documents," in *Proc. ICDAR Workshops*, 2019, pp. 1–6.

[14] S. Park, S. Shin, B. Lee, J. Lee, J. Surh, M. Seo, and H. Lee, "CORD: A consolidated receipt dataset for post-OCR parsing," in *NeurIPS 2019 Workshop on Document Intelligence*, 2019.

[15] Z. Huang, K. Chen, J. He, X. Bai, D. Karatzas, S. Lu, and C. V. Jawahar, "ICDAR2019 competition on scanned receipt OCR and information extraction," in *Proc. International Conference on Document Analysis and Recognition (ICDAR)*, 2019, pp. 1516–1520.

[16] B. Pfitzmann, C. Auer, M. Dolfi, A. S. Nassar, and P. Staar, "DocLayNet: A large human-annotated dataset for document-layout segmentation," in *Proc. 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2022, pp. 3743–3751.

[17] R. Smith, "An overview of the Tesseract OCR engine," in *Proc. 9th International Conference on Document Analysis and Recognition (ICDAR)*, 2007, pp. 629–633.

[18] K. Lee, M. Joshi, I. R. Turc, H. Hu, F. Liu, J. M. Eisenschlos, U. Khandelwal, P. Shaw, M.-W. Chang, and K. Toutanova, "Pix2Struct: Screenshot parsing as pretraining for visual language understanding," in *Proc. 40th International Conference on Machine Learning (ICML)*, PMLR vol. 202, 2023, pp. 18893–18912.

[19] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th International Conference on Machine Learning (ICML)*, PMLR vol. 70, 2017, pp. 1321–1330.

[20] M. Mathew, D. Karatzas, and C. V. Jawahar, "DocVQA: A dataset for VQA on document images," in *Proc. IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021, pp. 2199–2208.

[21] D. Hendrycks, C. Burns, A. Chen, and S. Ball, "CUAD: An expert-annotated NLP dataset for legal contract review," in *Proc. Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1, 2021.

[22] T. Stanisławek, F. Graliński, A. Wróblewska, D. Lipiński, A. Kaliska, P. Rosalska, B. Topolski, and P. Biecek, "Kleister: Key information extraction datasets involving long documents with complex layouts," in *Proc. International Conference on Document Analysis and Recognition (ICDAR)*, LNCS vol. 12821, 2021, pp. 564–579.

[23] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.

[24] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *Proc. International Conference on Learning Representations (ICLR)*, 2023.

[25] J. G. Fiscus, "A post-processing system to yield reduced word error rates: Recognizer output voting error reduction (ROVER)," in *Proc. IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU)*, 1997, pp. 347–354.

[26] P. Buneman, S. Khanna, and W.-C. Tan, "Why and where: A characterization of data provenance," in *Proc. 8th International Conference on Database Theory (ICDT)*, LNCS vol. 1973, 2001, pp. 316–330.

[27] E. Rahm and P. A. Bernstein, "A survey of approaches to automatic schema matching," *The VLDB Journal*, vol. 10, no. 4, pp. 334–350, 2001.

[28] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992.

[29] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, "Parameter-efficient transfer learning for NLP," in *Proc. 36th International Conference on Machine Learning (ICML)*, PMLR vol. 97, 2019, pp. 2790–2799.

[30] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proc. International Conference on Learning Representations (ICLR)*, 2022.